

Formulae

→ Green's Theorem

Circulation Curl.

$$\int F \cdot dr = \oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy.$$

$$\downarrow$$

$$\oint_C F \cdot T ds$$

and.

outward flux (Normal form).

$$\int F \cdot dr = \oint_C M dy - N dx = \iint_R \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \right) dx dy.$$

$$\downarrow$$

$$\oint_C F \cdot n ds.$$

→ Line Integral.

work done.

$$\text{work} = \int F \cdot dr.$$

general.

$$\int_C M dx + N dy + P dz.$$

→ Surface Integral

parameterization of surfaces.

$$r(u,v) = f(u,v)i + g(u,v)j + h(u,v)k.$$

→ Stoke's Theorem

$$\oint_C F \cdot dr = \iint_S (\nabla \times F) \cdot n d\sigma$$



Extreme Values

Minima $f_{xx} > 0$ and $f_{xx} \cdot f_{yy} - (f_{xy})^2 > 0$,

Maxima $f_{xx} < 0$ and $f_{xx} \cdot f_{yy} - (f_{xy})^2 > 0$,

Saddle point, if $f_{xx} \cdot f_{yy} - (f_{xy})^2 < 0$

Divergence Theorem.

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, d\sigma = \iiint_D \nabla \cdot \mathbf{F} \, dV.$$

Laplace Transform

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} e^{-st} f(t) \, dt.$$

Lagrange's Multiplier.

$$\nabla f = \lambda \nabla g.$$

Fourier Integral

$$f(x) = \int_0^{\infty} [A(\omega) \cos \omega x + B(\omega) \sin \omega x] \, d\omega$$

Fourier Periodic Series.

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right]$$

$$a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx.$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx.$$

Fourier Cosine Series.

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) \right]$$

Fourier Sine Series.

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left[b_n \sin\left(\frac{n\pi x}{L}\right) \right]$$

$$f(x) = \frac{1}{2} a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx, \quad a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cdot \cos nx \, dx.$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cdot \sin nx \, dx$$



Laplace Transform

$$F(s) = \mathcal{L}(f) = \int_0^{\infty} e^{-st} f(t) dt.$$

$$\mathcal{L}(1) = \frac{1}{s}$$

$$\mathcal{L}(t) = \frac{1}{s^2}$$

$$\mathcal{L}(t^2) = \frac{2!}{s^3}$$

$$\mathcal{L}(e^{at}) = \frac{1}{s-a}$$

$$\mathcal{L}(\sin at) = \frac{a}{s^2 + a^2}$$

$$\mathcal{L}(\sinh at) = \frac{a}{s^2 - a^2}$$

$$\mathcal{L}(\cos at) = \frac{s}{s^2 + a^2}$$

$$\mathcal{L}(\cosh at) = \frac{s}{s^2 - a^2}$$



6.1.

1) $3t + 12.$

$$\therefore \mathcal{L}(t) = \frac{1}{s^2}, \quad \mathcal{L}(1) = \frac{1}{s}$$

$$= 3 \mathcal{L}(t) + 12 \mathcal{L}(1)$$

$$= \frac{3}{s^2} + \frac{12}{s} \quad \text{-- Am.}$$

2) $(a-bt)^2$

$$= a^2 \mathcal{L}(1) - 2abt + b^2 t^2 \quad \because \{ (a-b)^2 = a^2 - 2ab + b^2. \}$$

$$= \frac{a^2}{s} - \frac{2ab}{s^2} + \frac{2b^2}{s^3}$$

-- Am.

$$\therefore t^2 = \frac{2!}{s^3} = \frac{2 \times 1}{s^3} = \frac{2}{s^3}$$

3) $\cos \pi t.$

$$a = \pi.$$

$$\therefore \cos at = \frac{s}{s^2 + a^2}$$

$$\text{So, } \frac{s}{s^2 + \pi^2} \quad \text{-- Am.}$$

4)



Find Laplace transform for $f(t) = \sin 2t \cos 3t$.

$$= \sin 2t \cos 3t$$

$$\because 2 \sin A \cos B = \sin(A+B) +$$

$$= \frac{1}{2} \left\{ 2 \sin 2t \cos 3t \right\}$$

$$\sin(A-B)$$

$$= \frac{1}{2} \left\{ \sin(2t+3t) + \sin(2t-3t) \right\}$$

$$= \frac{1}{2} \left\{ \sin 5t + \sin(-t) \right\}$$

$$\because \mathcal{L}(\sin at) = \frac{a}{s^2 + a^2}$$

$$= \frac{1}{2} \left\{ \mathcal{L}(\sin 5t) - \mathcal{L}(\sin t) \right\}$$

$$= \frac{1}{2} \left\{ \frac{5}{s^2 + 25} - \frac{1}{s^2 + 1} \right\}$$

$$= \frac{1}{2} \left\{ \frac{5s^2 + 5}{(s^2 + 25)(s^2 + 1)} - \frac{s^2 - 1}{(s^2 + 25)(s^2 + 1)} \right\}$$

$$= \frac{1}{2} \left\{ \frac{4s^2 - 20}{(s^2 + 25)(s^2 + 1)} \right\}$$

$$= \frac{1}{2} \cdot 4 \left\{ \frac{s^2 - 5}{(s^2 + 25)(s^2 + 1)} \right\}$$

$$= \frac{2(s^2 - 5)}{(s^2 + 25)(s^2 + 1)} \quad \text{--- Ans.}$$



Inverse Laplace

Find $L^{-1} \left[\frac{3s + 2}{s^2 - 9} \right]$

$$= L^{-1} \left[\frac{3s}{s^2 - 9} \right] + L^{-1} \left[\frac{2}{s^2 - 9} \right]$$

$$= 3 L^{-1} \left[\frac{s}{s^2 - 3^2} \right] + 2 L^{-1} \left[\frac{1}{s^2 - 3^2} \right] \quad \because \frac{s}{s^2 - a^2} = \cosh at$$

$$= 3 \cosh(3t) + 2 \cdot \frac{1}{3} \sinh 3t \quad \because \frac{1}{s^2 - a^2} = \frac{1}{a} \sinh at$$

$$= 3 \cosh(3t) + \frac{2}{3} \sinh(3t).$$

-- Ans.

G.1

30 $\rightarrow$ $\frac{4s + 32}{s^2 - 16}$

$$= L^{-1} \left[\frac{4s}{s^2 - 16} \right] + L^{-1} \left[\frac{32}{s^2 - 16} \right]$$

$$= 4 L^{-1} \left[\frac{s}{s^2 - 4^2} \right] + 8 L^{-1} \left[\frac{4}{s^2 - 4^2} \right]$$

$$= 4 L^{-1} \left[\cosh 4t \right] + 8 L^{-1} \left[\sinh 4t \right]$$

$$= 4 \cosh 4t + 8 \sinh 4t.$$

-- Ans.

$$\because \cosh at = \frac{s}{s^2 - a^2}$$

$$\because \sinh at = \frac{a}{s^2 - a^2}$$



Date _____

$$\therefore \cos at = \frac{s}{s^2 + a^2}$$

$$25) F(s) = \frac{0.2s + 1.4}{s^2 + 1.96}$$

$$\therefore \sin at = \frac{a}{s^2 + a^2}$$

$$= 0.2 \mathcal{L}^{-1} \left[\frac{s}{s^2 + (1.4)^2} \right] + \mathcal{L}^{-1} \left[\frac{1.4}{s^2 + (1.4)^2} \right]$$

$$= 0.2 (\cos 1.4t) + \sin 1.4t.$$

-- Ans.

$$27) F(s) = \frac{s}{L^2 s^2 + \frac{1}{4} \pi^2}$$

$$= \frac{1}{L^2} \mathcal{L}^{-1} \left[\frac{s}{s^2 + (\frac{1}{2}\pi)^2} \right]$$

$$\therefore \cos at = \frac{s}{s^2 + a^2}$$

$$= \frac{1}{L^2} \cos \left(\frac{1}{2}\pi \right) t.$$

-- Ans.



Laplace.

First Shifting Theorem (s - Shifting)

$$1 = \frac{1}{s} \Rightarrow \frac{1}{s-a}$$

Find Laplace transform of $e^{-at} \sin^2 t$.

$$\cos at = \frac{s}{s^2 + a^2} = \frac{s+a}{(s-a)^2 + a^2}$$

we will not consider e^{-at} .

$$= L \left\{ \sin^2 t \right\}$$

$$= L \left\{ \frac{1 - \cos 2t}{2} \right\}$$

$$= L \left\{ \frac{1}{2} \right\} - L \left\{ \frac{\cos 2t}{2} \right\}$$

$$= \frac{1}{2} L \{1\} - \frac{1}{2} L \{ \cos 2t \}$$

$$= \frac{1}{2(s-2)} - \frac{1}{2} \frac{(s+2)}{(s-2)^2 + 8}$$

$$= \frac{1}{2s-4} - \frac{(s+2)}{2s^2 + 8s + 16} \quad \text{--- Ans.}$$



Laplace S-Shifting.

33) $t^2 e^{-3t}$

ignore.

$$\frac{2!}{s^3} \Rightarrow \frac{2}{(s-3)^3} \quad \text{--- Ans.}$$

→

34) $k e^{-at} \cos \omega t$

L → ignore.

$$\cos at = \frac{s}{s^2 + a^2}$$

$$= k \cdot \left[\frac{s}{s^2 + \omega^2} \right]$$

$$= k \frac{(s+a)}{(s+a)^2 + \omega^2} \quad \text{--- Ans.}$$

35) $0.5 e^{-4.5t} \sin 2\pi t$

L → ignore.

$$= 0.5 \mathcal{L} \{ \sin 2\pi t \}$$

$$\sin at = \frac{a}{s^2 + a^2}$$

$$= 0.5 \frac{2\pi}{s^2 + (2\pi)^2}$$

$$= 0.5 \frac{2\pi}{(s+a)^2 + (2\pi)^2} \quad \text{--- Ans.}$$



Unit step function.

$$\textcircled{1} \quad \underline{\text{No shift}} \quad u(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$$

$$\mathcal{L}\{u(t)\} = \frac{1}{s}$$

$$\textcircled{2} \quad \underline{\text{with shift}} \quad u(t-a) = \begin{cases} 0 & t < a \\ 1 & t \geq a \end{cases}$$

$$\mathcal{L}\{u(t-a)\} = \frac{e^{-as}}{s}$$



$$t-3 \quad (t > 3).$$

unit function.

$$t > 3. \quad , \text{ So,}$$

$$f(t) = (t-a) u(t-a).$$

$$= (t-3) u(t-3).$$

Transform

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{(t-3) u(t-3)\}$$

$$\text{As we know, } \mathcal{L}\{f(t-a) u(t-a)\} = e^{-as} \mathcal{L}\{f(t)\}$$

$$e^{-3s} \mathcal{L}\{f(t)\}.$$

$$e^{-3s} \times \frac{1}{s^2} = \frac{e^{-3s}}{s^2}.$$

$$4) \cos 2t \quad (0 < t < \pi).$$

$$\text{unit step f: } f(t) = \cos 2t \left((t-0) u(t-0) - (t-\pi) u(t-\pi) \right)$$

$$= \cos 2t \left(u(t) - u(t-\pi) \right)$$

Transform the function.

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{\cos 2t u(t)\} - \mathcal{L}\{\cos 2t u(t-\pi)\}$$

$$= \frac{s}{s^2+4} - e^{-\pi s} \left(\frac{s}{s^2+4} \right)$$



$$e^{-t} \quad (0 < t < \pi).$$

unit step function.

$$f(t) = e^{-t} (u(t) - u(t - \pi)).$$

Transfer y.

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{e^{-t} u(t) - e^{-t} u(t - \pi)\}$$

$$= \frac{1}{s-1} - e^{-\pi s} \left(\frac{1}{s-1} \right)$$

